CO-2
ASSESSMENT TOOL 2

COURSE CODE: CSA16 COURSE NAME:Data Warehousing and Data Mining

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COURSE OUTCOME COVERED
CO2: Apply suitable Data Pre-processing techniques like Data cleaning, Data integration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)

CLASS TEST

Q1. Define Data Mining and list
any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it.
[L2, L4]

Q2. What is data cleaning? Mention
two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1,
L2, L3]

Q3. Define data integration and list
two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration. [L2,
L4]

Q4. What is covariance analysis? List the two covariance
formals[L2, L3]

Q5. Suppose two stocks A and B have the following
values in one week: (2, 5), (3, 8), (5,
10), (4, 11), (6, 14). Question: If the
stocks are affected by the same industry trends, will their prices rise or fall
together and independent ?

Q6. Explain the architecture of a
typical data mining system with a neat diagram. Discuss the role of each
component and analyze how they interact to support knowledge discovery. [L2, L4]

Q7. Find covariance for following data set x = {4,5,6,7,9}, y ={8,3,7,5,6}

Q8. Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps. [L3, L4, L5]

Q9. A hospital is analyzing patient **age** data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency** (equi-depth) binning. **Sample Data Set (Ages):** 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]

Q10. Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples, and evaluate their impact on data analysis. [L3, L4, l5]

Answers:

1) **Data Mining:**
Data Mining is the process of discovering useful, meaningful, and previously unknown patterns, relationships, and knowledge from large volumes of data using statistical, machine learning, and database techniques.

Two Basic DataMiningTasks

1. **Classification** – Assigns data into predefined categories.
   - Example: Classifying emails as Spam or Not Spam.
2. **Clustering** – Groups similar data objects without predefined labels.
   - Example: Grouping customers based on purchasing behavior.

Steps in the KDD (Knowledge Discovery in Databases) Process

1. **Data Cleaning**
   - Removesmissing values, duplicate records, and noise.
   - Example:Filling missing customer age values.
2. **Data Integration**
   - Combinesdata from multiple sources into one dataset.
   - Example:Merging sales and customer databases.
3. **Data Selection**
   - Selectsonlyrelevant data required for analysis.
   - Example:Selecting customer purchase records for the last year.
4. **Data Transformation**
   - Convertsdata into suitable formats using normalization or aggregation.
   - Example:Scaling income values between 0 and 1.
5. **DataMining**
   - Appliesalgorithms to discover useful patterns.
   - Example:Finding products frequently bought together.

6. **PatternEvaluation**
    - Identifies interesting and useful patterns.
    - Example: Selecting association rules with high confidence.
7. **Knowledge Presentation**
    - Displays discovered knowledge using reports, graphs, or dashboards.

RoleofData Preprocessing
Data preprocessing improves data quality before mining.

Importance:
- Removes noise and inconsistencies.
- Handles missing values.
- Reduces redundancy.
- Improves mining accuracy.
- Reduces computational time.

Without preprocessing, data mining algorithms may produce inaccurate or misleading results. Clean and consistent data leads to reliable knowledge discovery.

2) Data cleaning is the process of detecting and correcting inaccurate, incomplete, duplicate, or inconsistent data.

TwoCommonDataQualityIssues
1. Missing values
2. Noisy (incorrect) data

Handling Missing Values
Methods
1. Ignore the record
2. Fill manually
3. Replace with mean/median/mode
4. Predict using machine learning

Example

Age
20
22
?
24

Mean = (20+22+24)/3 = **22**
Replace missing value with **22**.
Handling Noisy Data
Methods

- Binning
- Regression
- Clustering
- Outlier detection

Example
Original:
15, 16, 16, 18, **90**
90 is an outlier.
After removing:
15,16,16,18

3) Data Integration is the process of combining data from multiple sources into one unified view.

Challenges

1. Data redundancy
2. Data inconsistency

Handling Redundancy

Redundancy occurs when duplicate information exists.

Methods

- Duplicate detection
- Correlation analysis
- Schema matching

Example

CustomerID 101 appears in two databases.
Keep only one record.

Handling Inconsistency

Occurs when different values represent the same information.
Example:
Database A:
Male
Database B:
M
Convert both into a standard format (Male).
Other methods:

- Data standardization
- Data validation
- Constraint checking

4) Covariance analysis measures how two variables change together.

- Positive covariance → variables increase together.
- Negative covariance → one increases while the other decreases.
- Zero covariance → independent relationship.

Population Covariance

$$Cov(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

Sample Covariance

$$Cov(X,Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

5) Given:
A = 2,3,5,4,6
B = 5,8,10,11,14

Mean

$$\bar{x}=4$$
$$\bar{y}=9.6$$

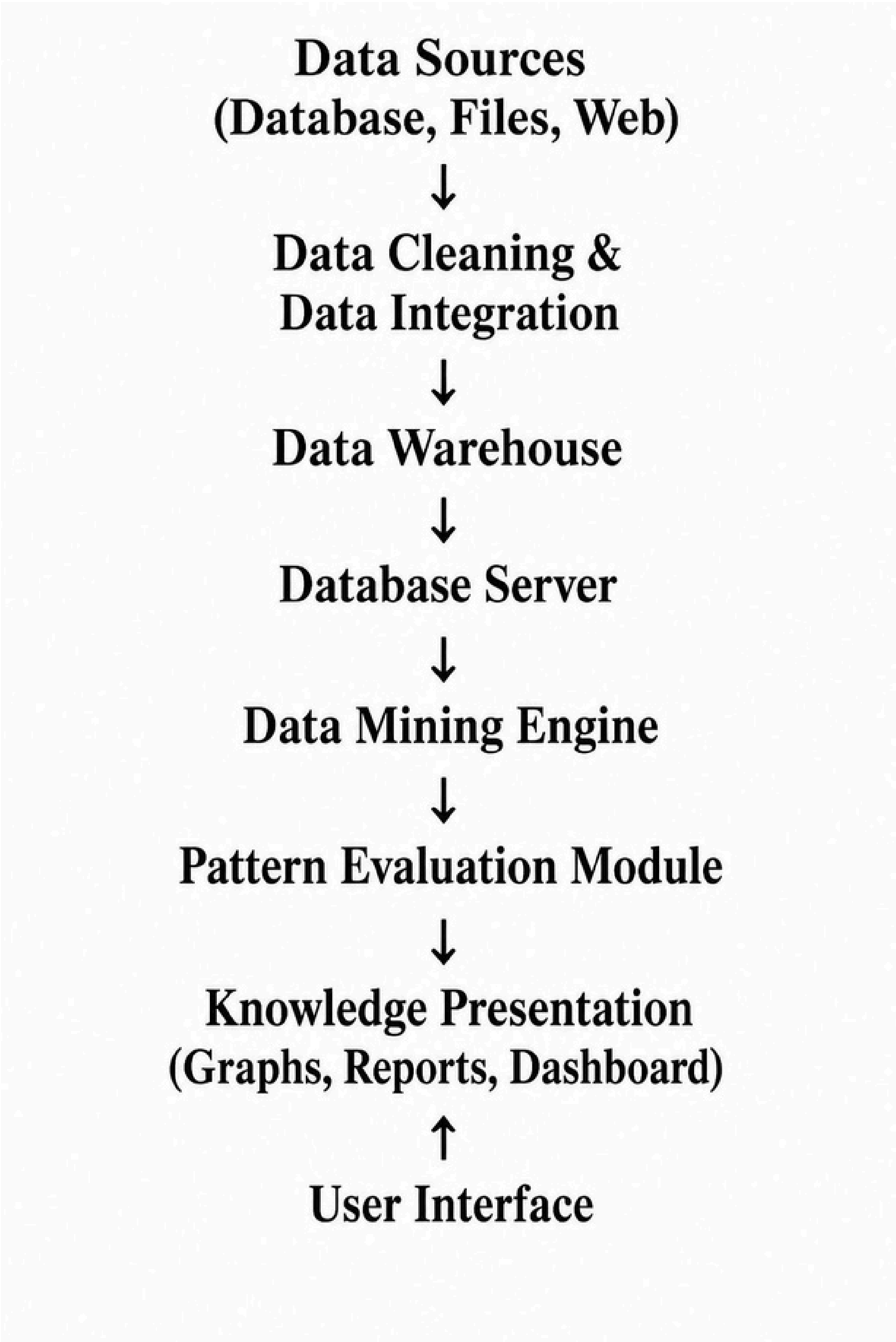
X	Y	X−4	Y−9.6	Product
2	5	-2	-4.6	9.2
3	8	-1	-1.6	1.6
5	10	1	0.4	0.4
4	11	0	1.4	0
6	14	2	4.4	8.8

Total = **20**
Sample covariance:

$$\frac{20}{4} = 5$$

Covariance = **5 (Positive)**

6)



7) Given:
x = {4,5,6,7,9}
y = {8,3,7,5,6}
Means

$$\bar{x}=6.2$$
$$\bar{y}=5.8$$

X	Y	X−6.2	Y−5.8	Product
4	8	-2.2	2.2	-4.84
5	3	-1.2	-2.8	3.36
6	7	-0.2	1.2	-0.24
7	5	0.8	-0.8	-0.64
9	6	2.8	0.2	0.56

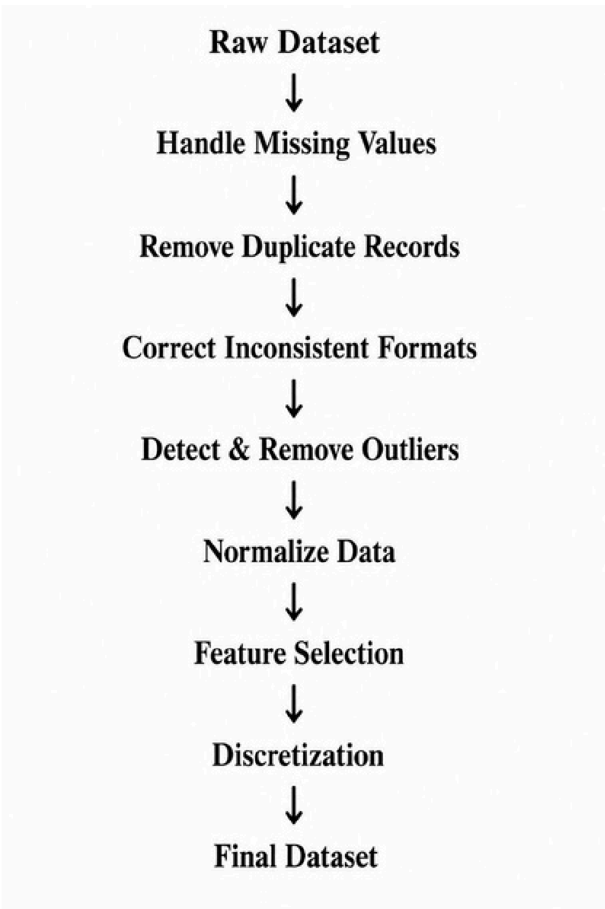
Total = **−1.8**
Sample covariance:

$$\frac{-1.8}{4} = -0.45$$

Covariance = **−0.45**

8) **Major Steps**

1. Data Cleaning
2. Data Integration
3. Data Transformation
4. Data Reduction
5. Data Discretization



Justification

- Missing values handled first.
- Duplicates removed before transformation.
- Standardize formats before analysis.
- Outliers removed before normalization.
- Feature selection reduces computation.
- Discretization prepares data for mining.

9) Data: 13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,46,52,70
Total = **27 values**
Choose **3 equal-frequency bins**
Each bin contains **9 values**.

Bin 1
13,15,16,16,19,20,20,21,22
Mean = **18**
Smoothed:
18,18,18,18,18,18,18,18,18

Bin 2
22,25,25,25,25,30,33,33,35
Mean = **28**
Smoothed:
28,28,28,28,28,28,28,28,28

Bin 3
35,35,35,36,40,45,46,52,70
Mean = **43**
Smoothed:
43,43,43,43,43,43,43,43,43

So, the data becomes smoother by replacing values in each bin with the bin mean, reducing the effect of minor variations and noise.

10) Data discretization converts continuous numerical values into a finite number of intervals (bins).

Importance

- Reduces noise.
- Simplifies data analysis.
- Improves algorithm performance.
- Reduces storage.
- Helps in classification and rule mining.

Equal-Width Binning
Formula

$$\text{Bin Width} = \frac{\text{Maximum} - \text{Minimum}}{\text{Number of Bins}}$$

Example
Data: 10,20,30,40,50,60
3 bins

Width

$$= \frac{60 - 10}{3} = 16.67$$

Bins:

- 10–26.67
- 26.67–43.34
- 43.34–60

Equal-Frequency Binning

Each bin contains the same number of records.
Example:
10,20,30,40,50,60
3 bins

- (10,20)
- (30,40)
- (50,60)

Each bin has **2values** .

Feature	**Equal-Width**	**Equal-Frequency**
Basis	Equal interval width	Equal number of records
Bin Size	Fixed width	Variable width
Data Distribution	May be uneven	More balanced
Sensitive to Outliers	Yes	Less sensitive
Best For	Uniformly distributed data	Skewed data

Impact on Data Analysis

- **Equal-widthbinning** is simple and suitable for uniformly distributed data but can create sparse bins when data is skewed.
- **Equal-frequencybinning** distributes records evenly across bins, making it more effective for skewed datasets and reducing the impact of outliers, though interval widths may vary.